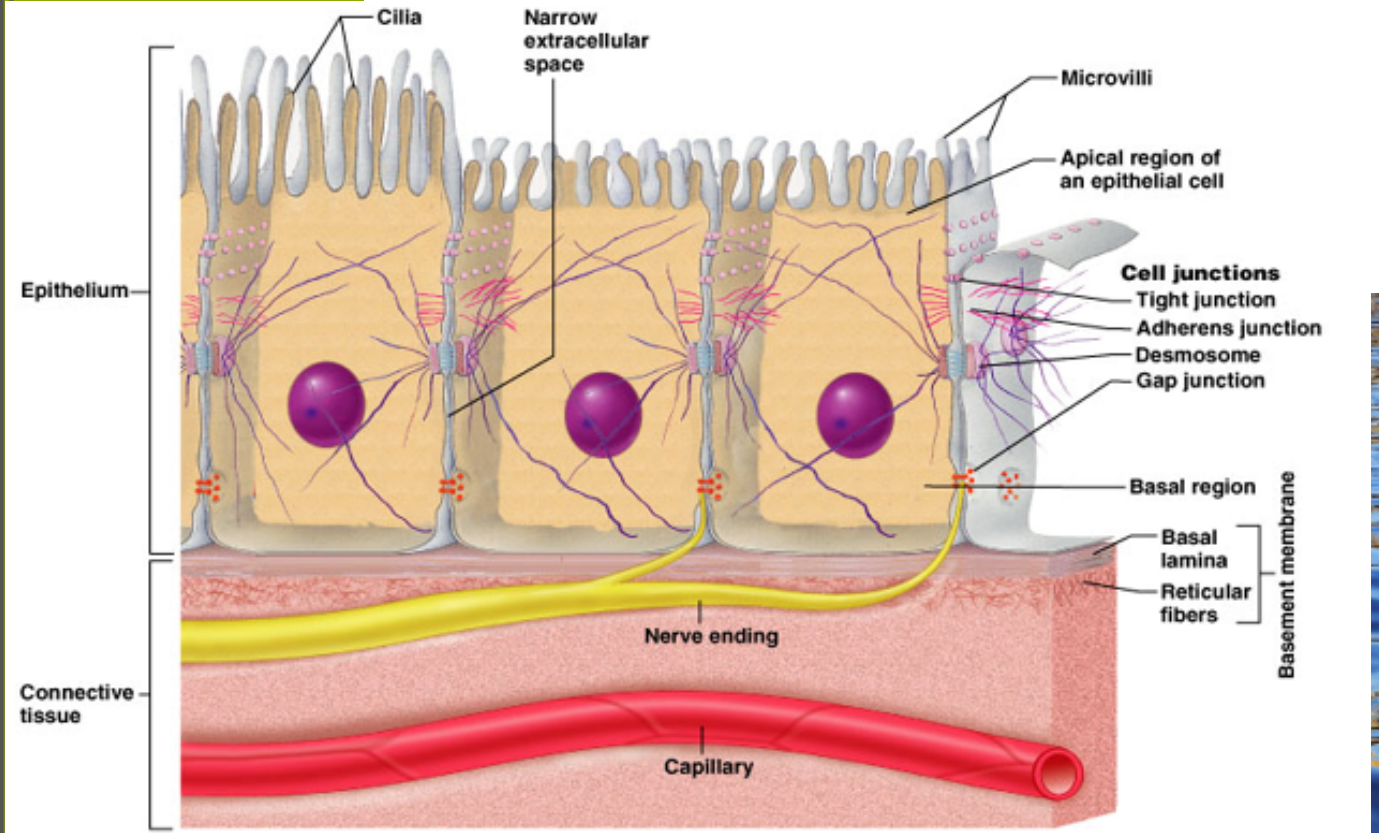
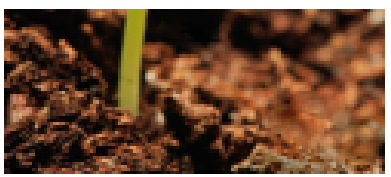
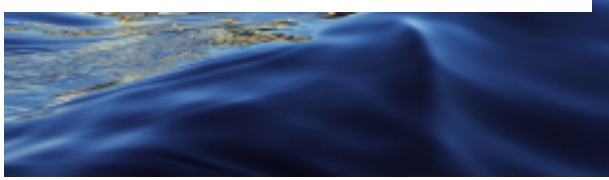
Organization Of Body Tissue

Prof.Dr. Mahmoud M. Ewidea







introduction

- Similar cells with a common function are called tissues.
- The study of tissues is called histology.
- There are four (4) primary or major tissue types:

1. Epithelial Tissue
2. Connective Tissue
3. Muscle Tissue
4. Nervous Tissue

The human body composition

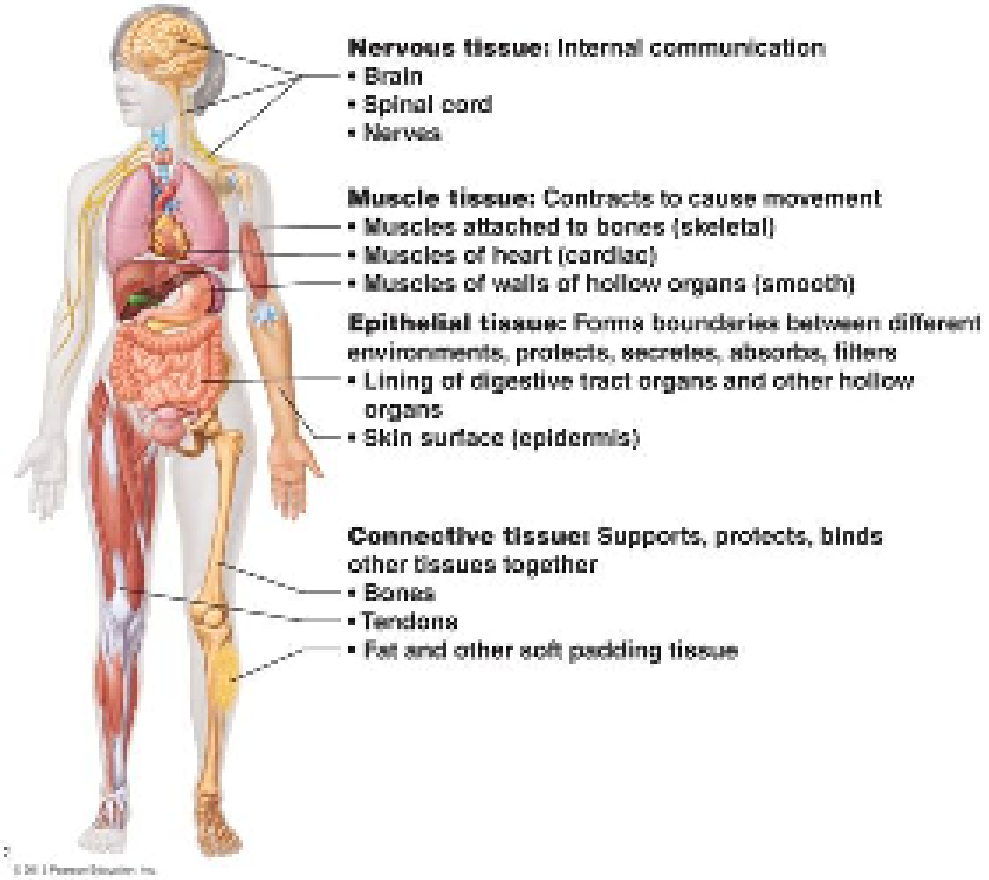
1. Epithelium: lines and covers surfaces
2. Connective tissue: protects, support, and bind together.
3. Muscular tissue: produces movement.
4. Nervous tissue: receives stimuli and conduct impulses .

1. Epithelial Tissue

A. Covers body surfaces

B. Lines body cavities and ducts

C. Forms glands

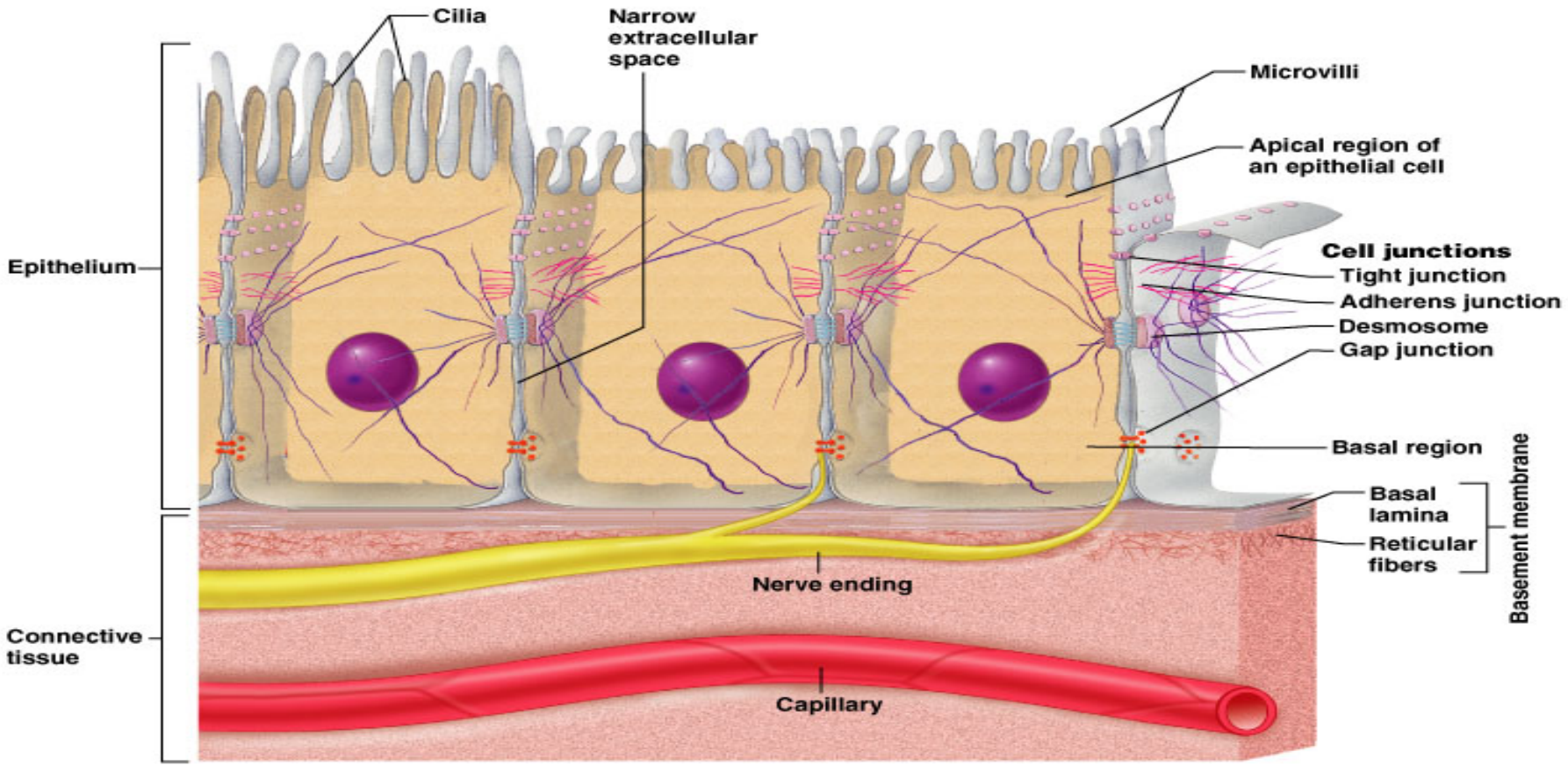


YOUR VISIBLE SKIN,
LINING OF GUT,
LINING OF BLOOD VESSELS,
RESPIRATORY PASSAGES,
DUCTS, JUST ABOUT ANY SURFACE AREA-NOSE, EAR, EYE

Special Characteristics of Epithelia

- Cellularity
  - cells are in close contact with each other with little or no intercellular space between them
- Specialized contacts
  - may have junctions for both attachment and communication
- Polarity
  - epithelial tissues always have an apical and basal surface
- Support by connective tissue
  - at the basal surface, both the epithelial tissue and the connective tissue contribute to the basement membrane
- Avascular
  - nutrients must diffuse from basal layer
- Innervated
- Regenerative
  - epithelial tissues are highly mitotic

Special Characteristics of Epithelia

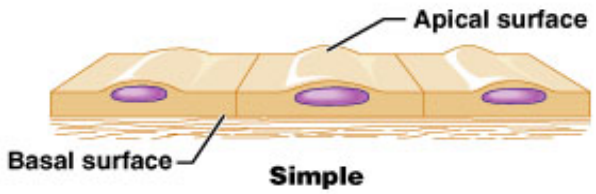


Epithelium cells

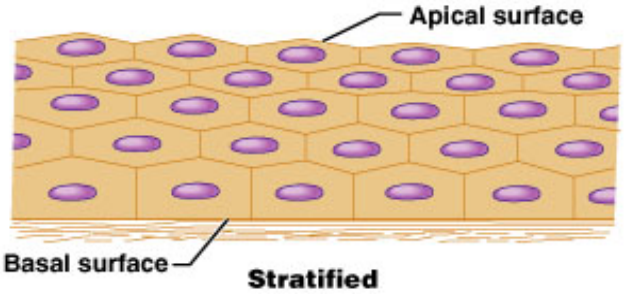
- Cells in epithelium fit closely together side by side and sometimes atop each other to form sheets of cells.
- These sheets are held together by specialized junctions. Supported by connective tissue.

Classifications of Epithelia

- First name of tissue indicates number of layers
  - Simple – one layer of cells

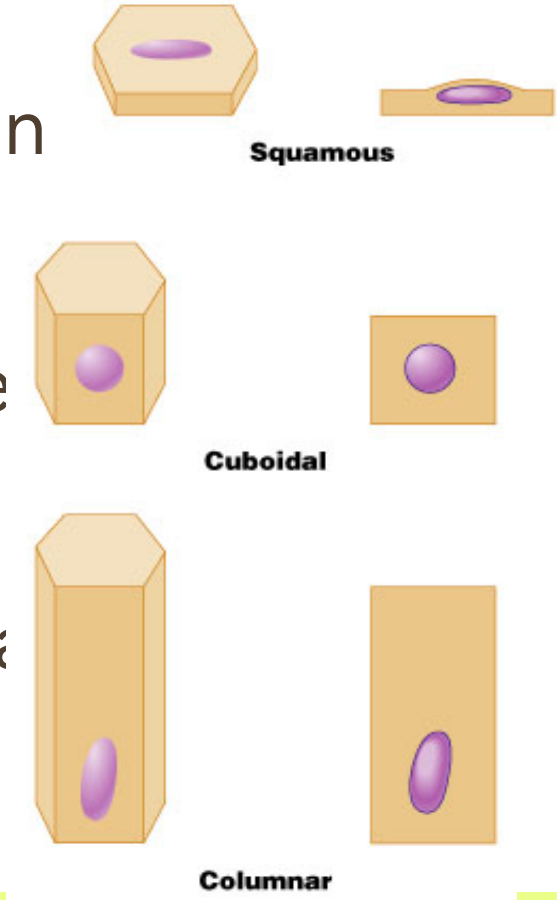


  - Stratified – more than one layer of cells



Classifications of Epithelia

- Last name of tissue describes shape of cells
  - Squamous – cells wider than tall (plate or “scale” like)
  - Cuboidal – cells are as wide tall, as in cubes
  - Columnar – cells are taller tha they are wide, like columns

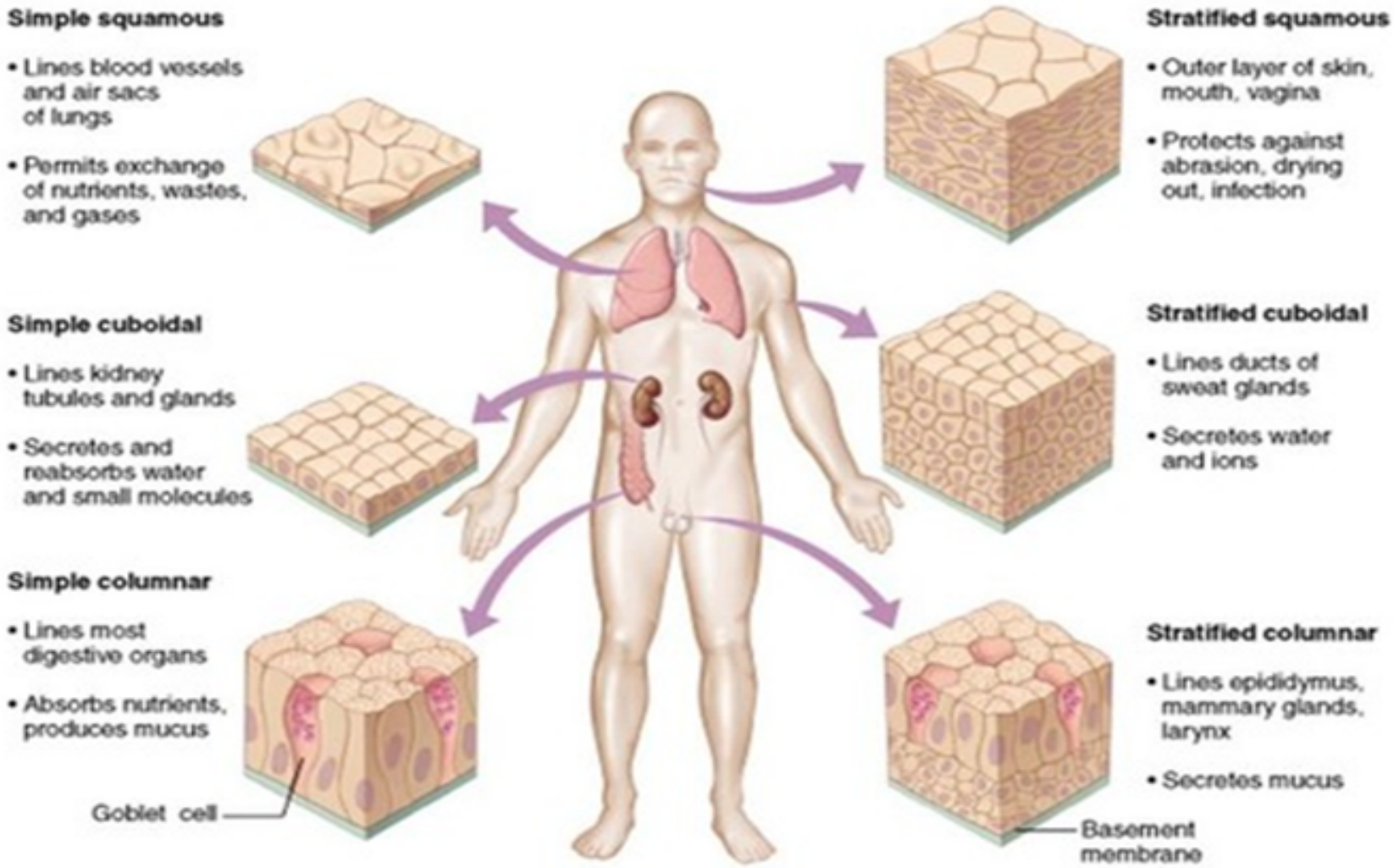


Naming Epithelia

- Naming the epithelia includes both the layers (first) and the shape of the cells (second)
  - i.e. stratified cuboidal epithelium
- The name may also include any accessory structures
  - Goblet cells
  - Cilia
  - Keratin
- Special epithelial tissues (don’t follow naming convention)
  - Psuedostratified
  - Transitional

Epithelium

- Attachment to a layer of connective tissue at the **basal surface** forms a layer called the **basement membrane**,
- **basement membrane** an adhesive layer formed by secretions from the epithelial cells and the connective tissue cells.
- **Avascular**– Epithelium typically lacks its own blood supply. Regeneration– Epithelium cells can regenerate if proper nourished.

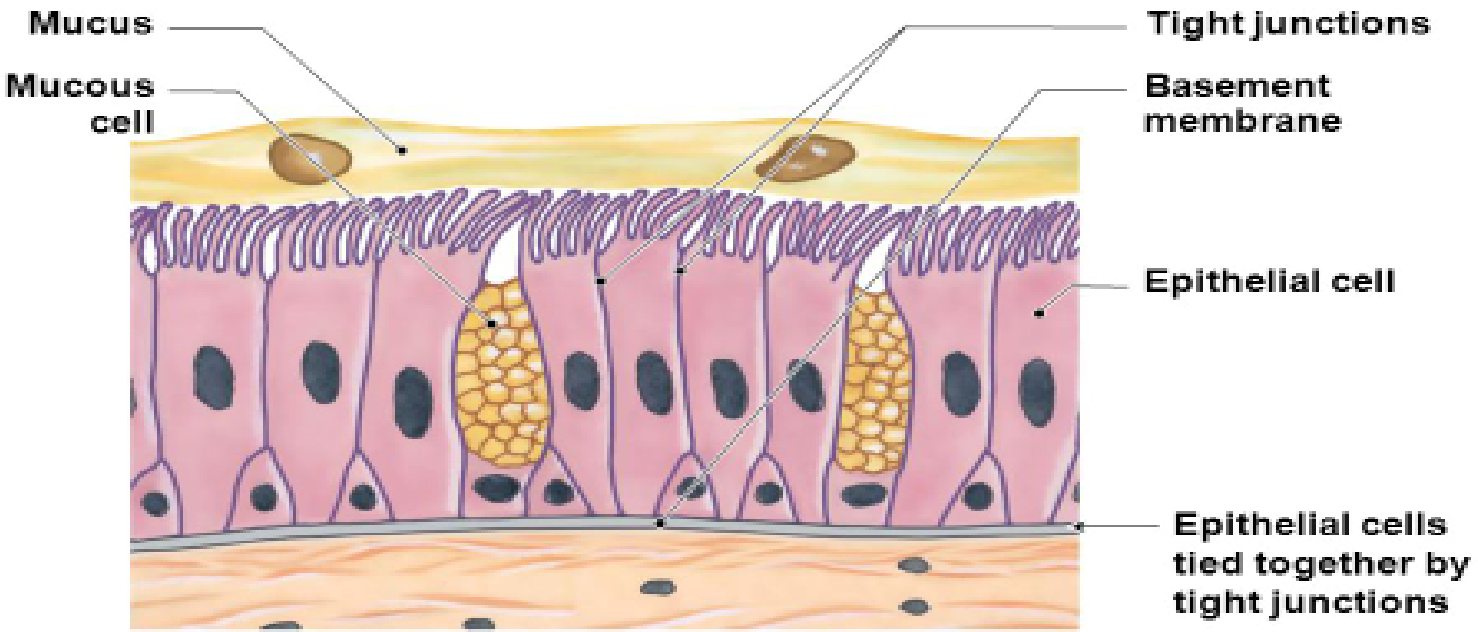


(a) Most epithelial tissues line or cover surfaces or body cavities

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basement membrane

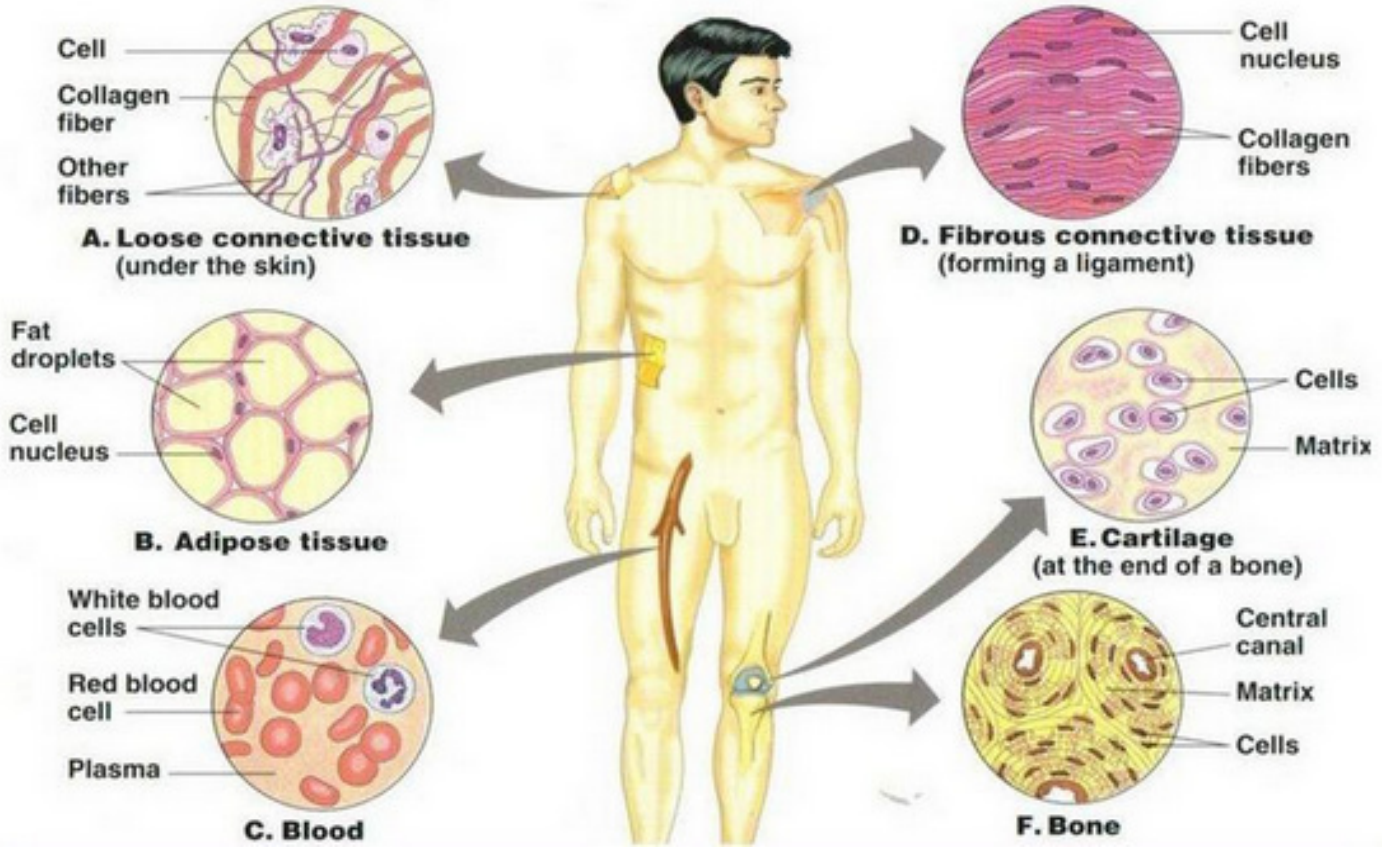
Mucous membranes as a nonspecific defense barrier



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Figure 13.5 2

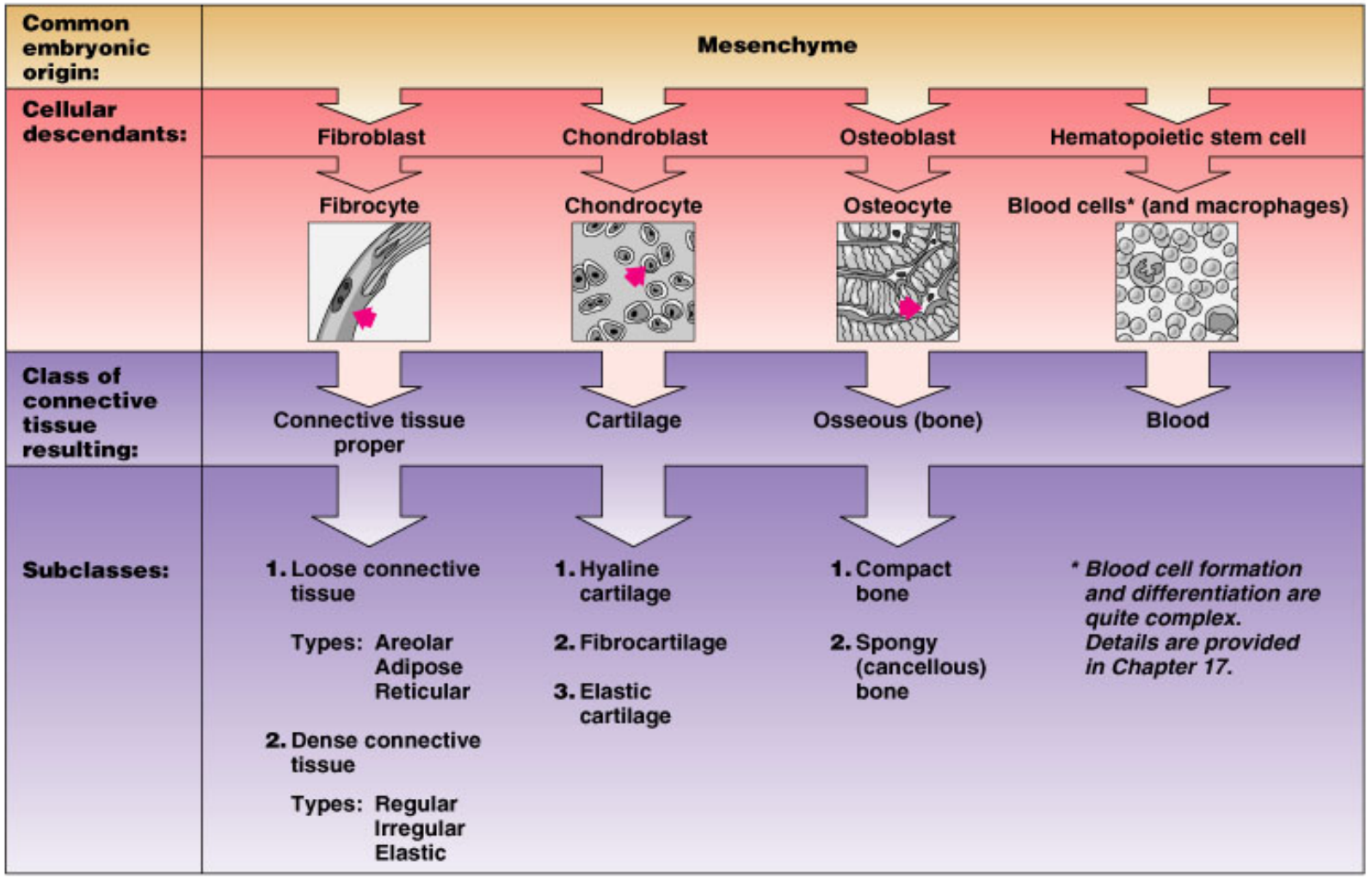
Connective tissue



Connective Tissue

- Most varied and abundant tissue
- Main classes
  - Connective tissue proper
  - Cartilage
  - Bone tissue
  - Blood
- Components of connective tissue:
  - Cells (varies according to tissue)
  - Matrix
    - Fibers (varies according to tissue)
    - Ground substance (varies according to tissue)
      - dermatin sulfate, hyaluronic acid, keratin sulfate, chondroitin sulfate…
- Common embryonic origin – mesenchyme

Classes of Connective Tissue



Connective Tissue Model

- Areolar connective tissue
  - Underlies epithelial tissue
  - Surrounds small nerves and blood vessels
  - Has structures and functions shared by other connective tissues
  - Borders all other tissues in the body
- Structures within areolar connective tissue allow:
  - Support and binding of other tissues
  - Holding body fluids
  - Defending body against infection
  - Storing nutrients as fat

Connective Tissue Proper

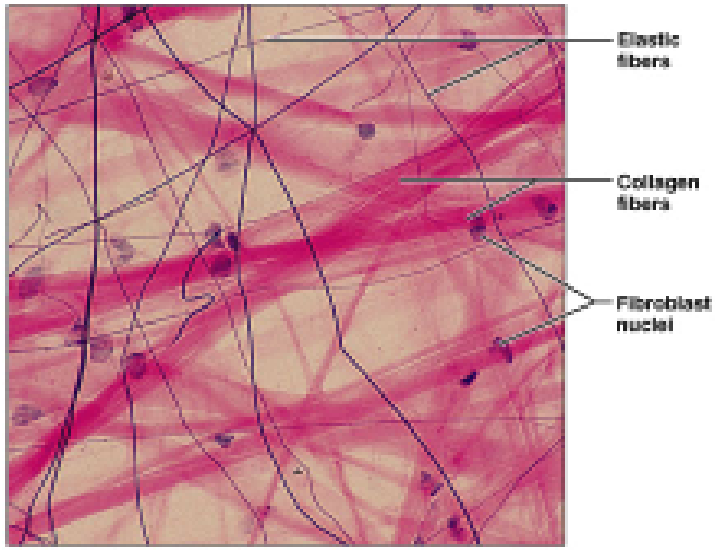
- Loose Connective Tissue
  - Areolar
  - Reticular
  - Adipose
- Dense Connective Tissue
  - Regular
  - Irregular
  - Elastic

Areolar Connective Tissue

- Description
  - Gel-like matrix with:
    - all three fiber types (collagen, reticular, elastic) for support
    - Ground substance is made up by glycoproteins also made and scretedby the fibroblasts.
  - Cells – fibroblasts, macrophages, mast cells, white blood cells
- Function
  - Wraps and cushions organs
  - Holds and conveys tissue fluid
  - Important role in inflammation
  - Main battlefield in fight against infection
- Defenders gather at infection sites
  - Macrophages
  - Plasma cells
  - Mast cells
  - Neutrophils, lymphocytes, and eosinophils

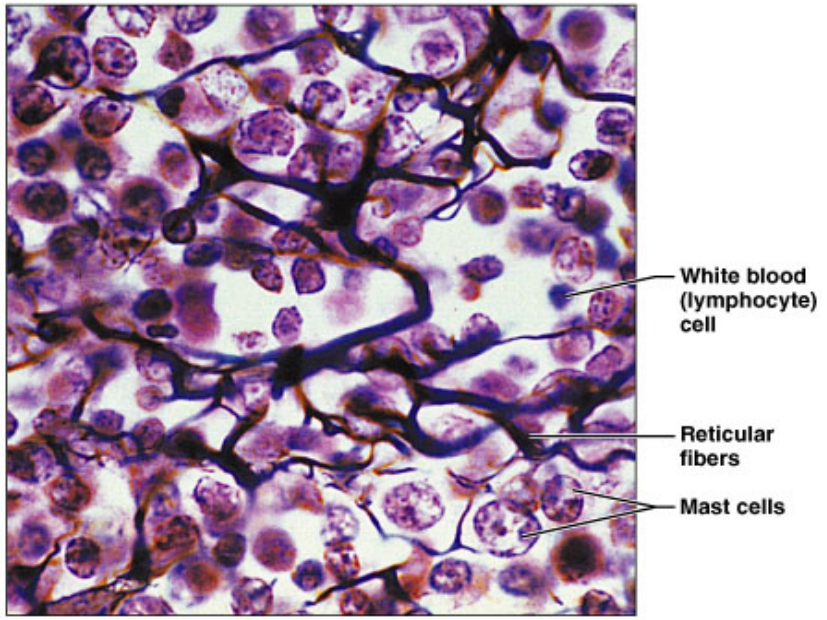
Areolar Connective Tissue

- Location
  - Widely distributed under epithelia
  - Packages organs
  - Surrounds capillaries



Reticular Connective Tissue

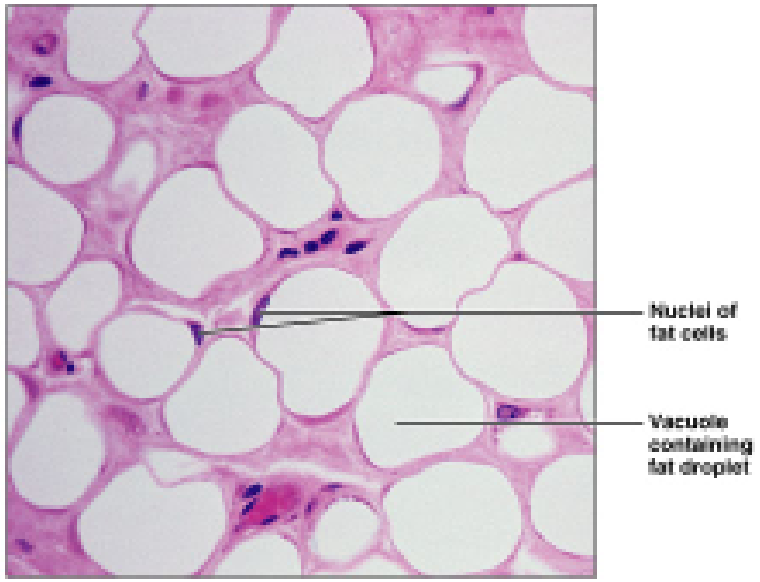
- Description – network of reticular fibers in loose ground substance
- Function – form a soft, internal skeleton (stroma) – supports other cell types
- Location – lymphoid organs
  - Lymph nodes, bone marrow, and spleen



Photomicrograph: Dark-staining network of reticular connective tissue fibers forming the internal skeleton of the spleen (350×).

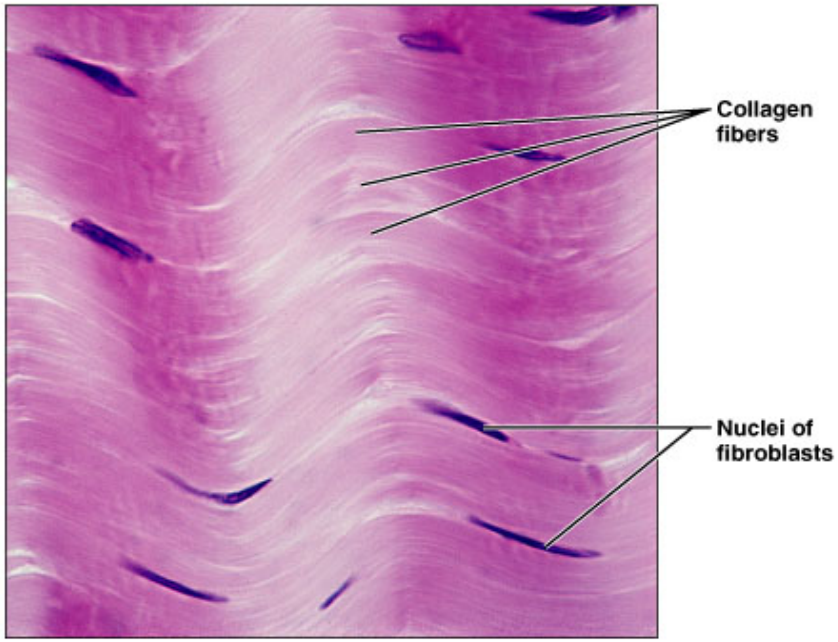
Adipose Tissue

- Description
  - Closely packed adipocytes
  - Have nucleus pushed to one side by fat droplet Function
  - Provides reserve food fuel
  - Insulates against heat loss
  - Supports and protects organs
- Location
  - Under skin
  - Around kidneys
  - Behind eyeballs, within abdomen and in breasts



Dense Regular Connective Tissue

- Description
  - Primarily *parallel* collagen fibers
  - Fibroblasts and some elastic fibers
  - Poorly vascularized
- Function
  - Attaches muscle to bon
  - Attaches bone to bone
  - Withstands great stress one direction
- Location
  - Tendons and ligaments
  - Aponeuroses
  - Fascia around muscles



Cartilage

- Characteristics:
  - Firm, flexible tissue
  - Contains no blood vessels or nerves
  - Matrix contains up to 80% water
  - Cell type – chondrocyte
- Types:
  - Hyaline
  - Elastic
  - Fibrocartilage

Hyaline Cartilage

- Description
  - Undetectable collagen fibers (hyaline = glassy)
  - Chondroblasts produce matrix
  - Chondrocytes lie in holes
- Function
  - Supports and reinforces
  - Strong cushion
  - Resists repetitive stress

Hyaline Cartilage

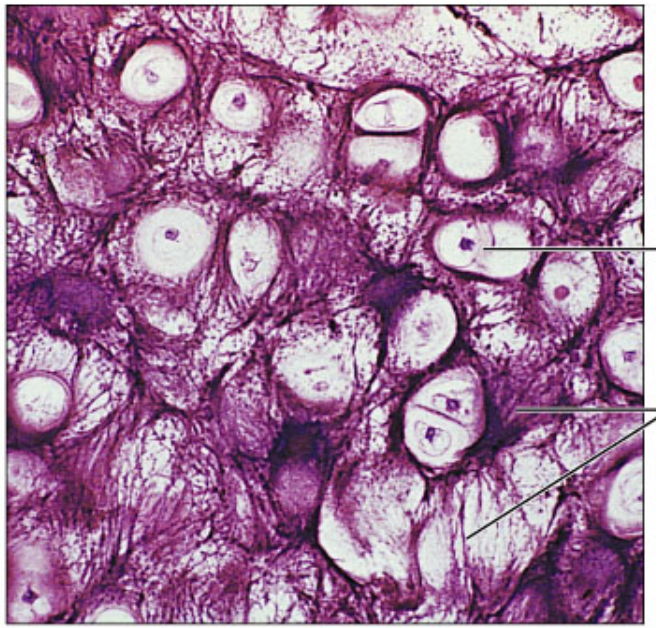
- Location
  - Fetal skeleton
  - Ends of long bones
  - Costal cartilage of r
  - Cartilages of nose, trachea, and larynx



Photomicrograph: Hyaline cartilage from the trachea (300×).

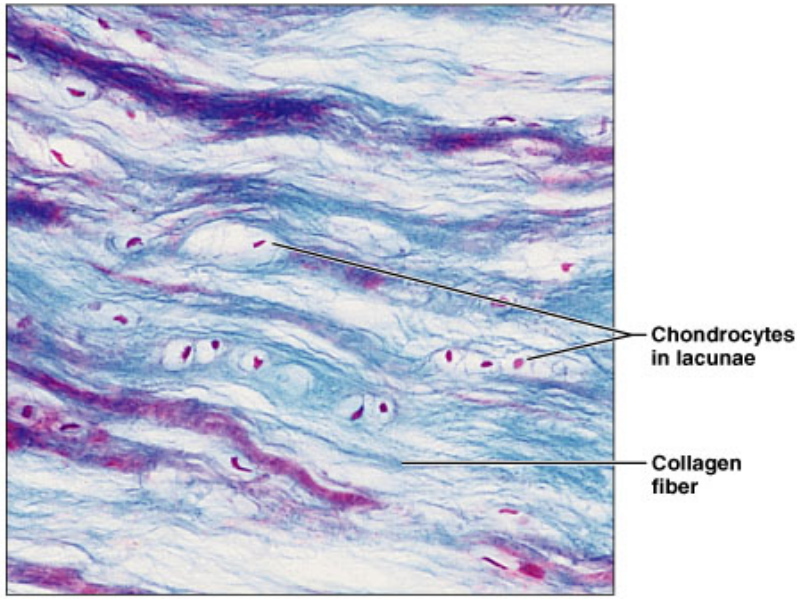
Elastic Cartilage

- Description
  - Like hyaline cartilage
  - More elastic fibers in m
- Function
  - Maintains shape of stru
  - Allows great flexibility
- Location
  - Supports external ear
  - Epiglottis



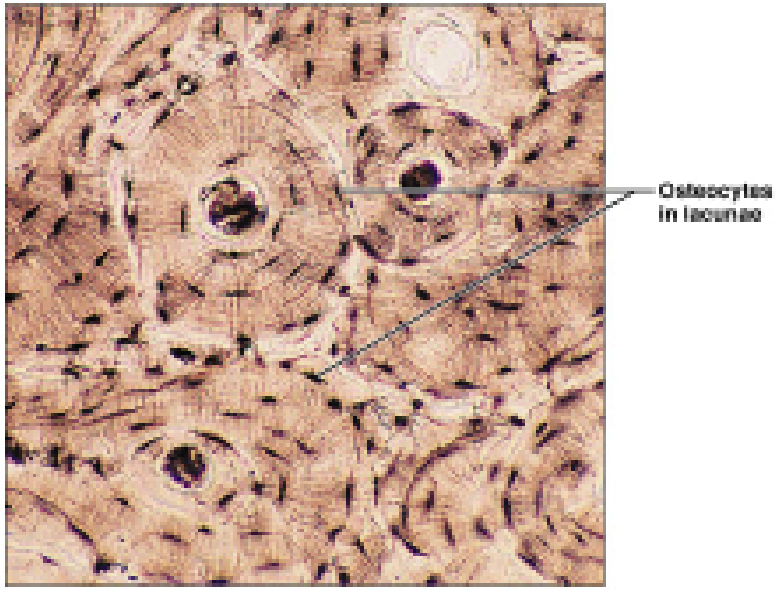
Fibrocartilage

- Description
  - Matrix similar, but less firm than hyaline cartilage
  - Thick collagen fibers predominate
- Function
  - Tensile strength and ability to absorb compressive shock
- Location
  - Intervertebral discs
  - Pubic symphysis
  - Discs of knee joint



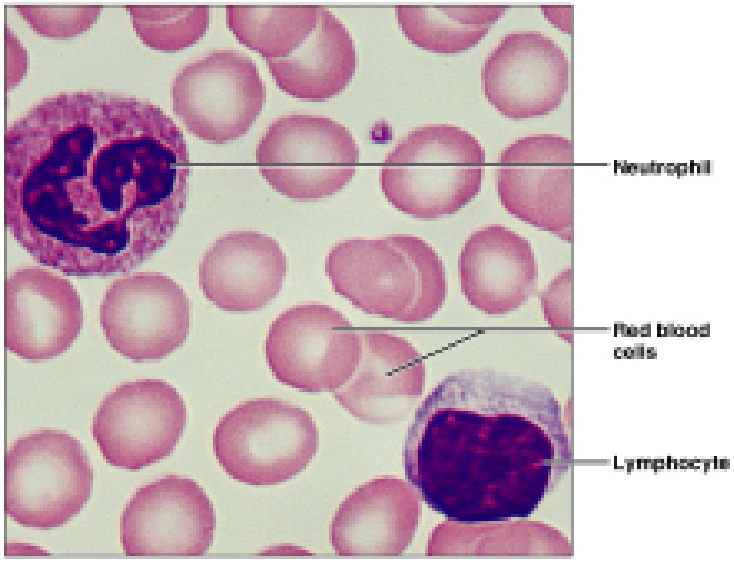
Bone Tissue

- Function
  - Supports and protects organs
  - Provides levers and attachment site for muscles
  - Stores calcium and other minerals
  - Stores fat
  - Marrow is site for blood cell formation
- Location
  - Bones



Blood Tissue

- Description
  - red and white blood cells in a fluid matrix
- Function
  - transport of respiratory gases, nutrients, and wastes
- Location
  - within blood vessels
- Characteristics
  - An atypical connective tissue
  - Develops from mesenchyme
  - Consists of cells surrounded by nonliving matrix



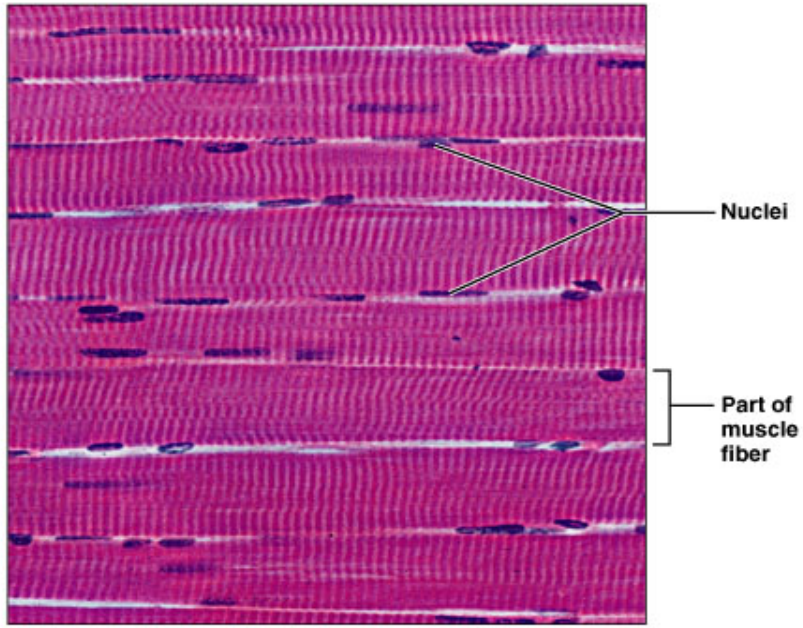
Photomicrograph: Smear of human blood (1500×); two white blood cells (neutrophil in upper left and lymphocyte in lower right) are seen surrounded by red blood cells.

Muscle Tissue

- Types
  - Skeletal muscle tissue
  - Cardiac muscle tissue
  - Smooth muscle tissue

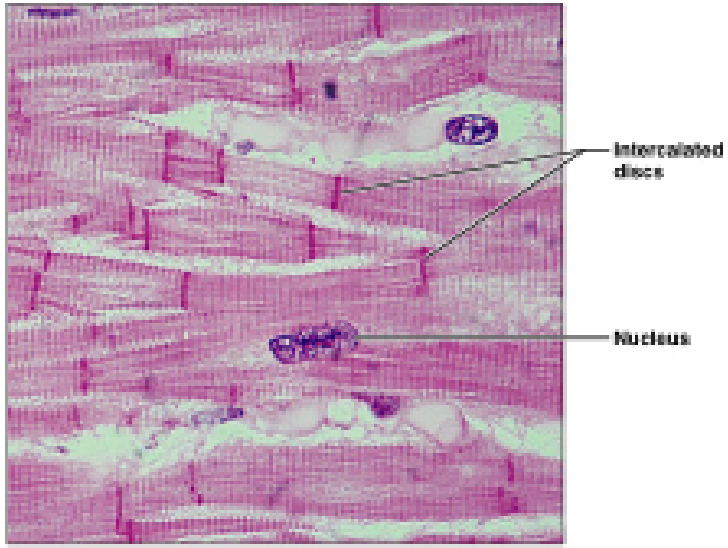
Skeletal Muscle Tissue

- Characteristics
  - Long, cylindrical cells
  - Multinucleate
  - Obvious striations
- Function
  - Voluntary movement
  - Manipulation of environment
  - Facial expression
- Location
  - Skeletal muscles attached to bones (occasionally to skin)



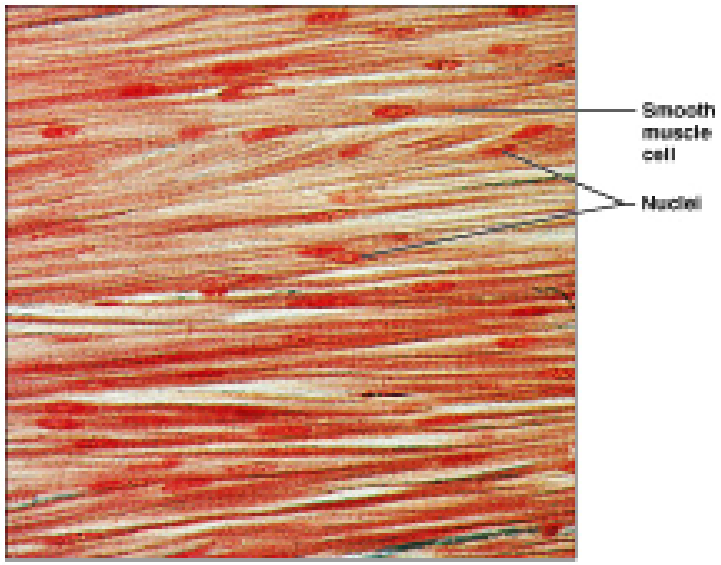
Cardiac Muscle Tissue

- Function
  - Contracts to propel blood into circulatory system
- Characteristics
  - Branching cells
  - Uni-nucleate
  - Striations
  - Intercalated discs
- Location
  - Occurs in walls of heart



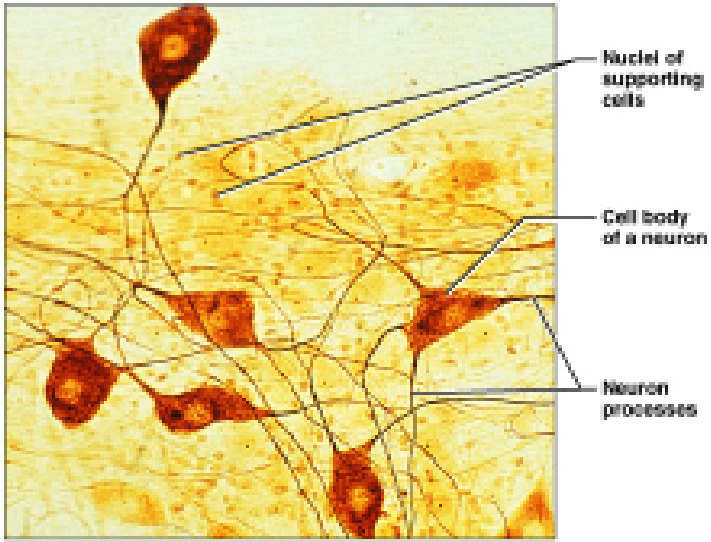
Smooth Muscle Tissue

- Characteristics
  - Spindle-shaped cells with central nuclei
  - Arranged closely to form sheets
  - No striations
- Function
  - Propels substances along internal passageways
  - Involuntary control
- Location
  - Mostly walls of hollow organs



Nervous Tissue

- Function
  - Transmit electrical signals from sensory receptors to effectors
- Location
  - Brain, spinal cord, and nerves
- Description
  - Main components are brain, spinal cord, and nerves
  - Contains two types of cells
    - Neurons – excitatory cells
    - Supporting cells (neuroglial cells)



Photomicrograph: Neurons (100×)

Tissue Response to Injury

- Inflammatory response – non-specific, local response
  - Limits damage to injury site
- Immune response – takes longer to develop and very specific
  - Destroys particular microorganisms at site of infection

The Tissues Throughout Life

- At the end of second month of development:
  - Primary tissue types have appeared
  - Major organs are in place
- Adulthood
  - Only a few tissues regenerate
  - Many tissues still retain populations of stem cells
- With increasing age:
  - Epithelia thin
  - Collagen decreases
  - Bones, muscles, and nervous tissue begin to atrophy
  - Poor nutrition and poor circulation – poor health of tissues

Functions of the skin

- Provides a protective barrier against mechanical, thermal and physical injury and hazardous substances.
- Prevents loss of moisture.
- Reduces harmful effects of UV radiation.
- Acts as a sensory organ (touch, detects temperature).
- Helps regulate temperature.
- An immune organ to detect infections etc.
- Production of vitamin D.

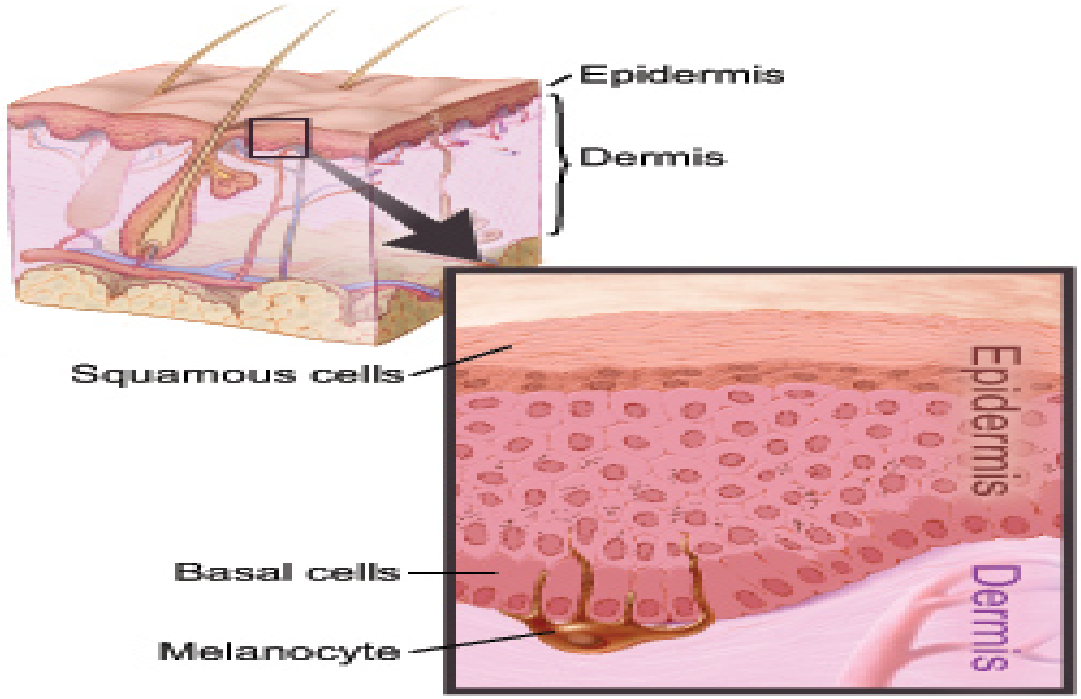
The epidermis

- The outer surface of the skin is the epidermis, which itself contains several layers — the basal cell layer, the spinous cell layer, the granular cell layer, and the stratum corneum. The cells in the epidermis are called keratinocytes.
- The deepest layer of the epidermis is the basal cell layer. Here cells are continually dividing to produce plump new skin cells (millions daily). These cells move towards the skin surface, pushed upward by the dividing cells below them.

The epidermis

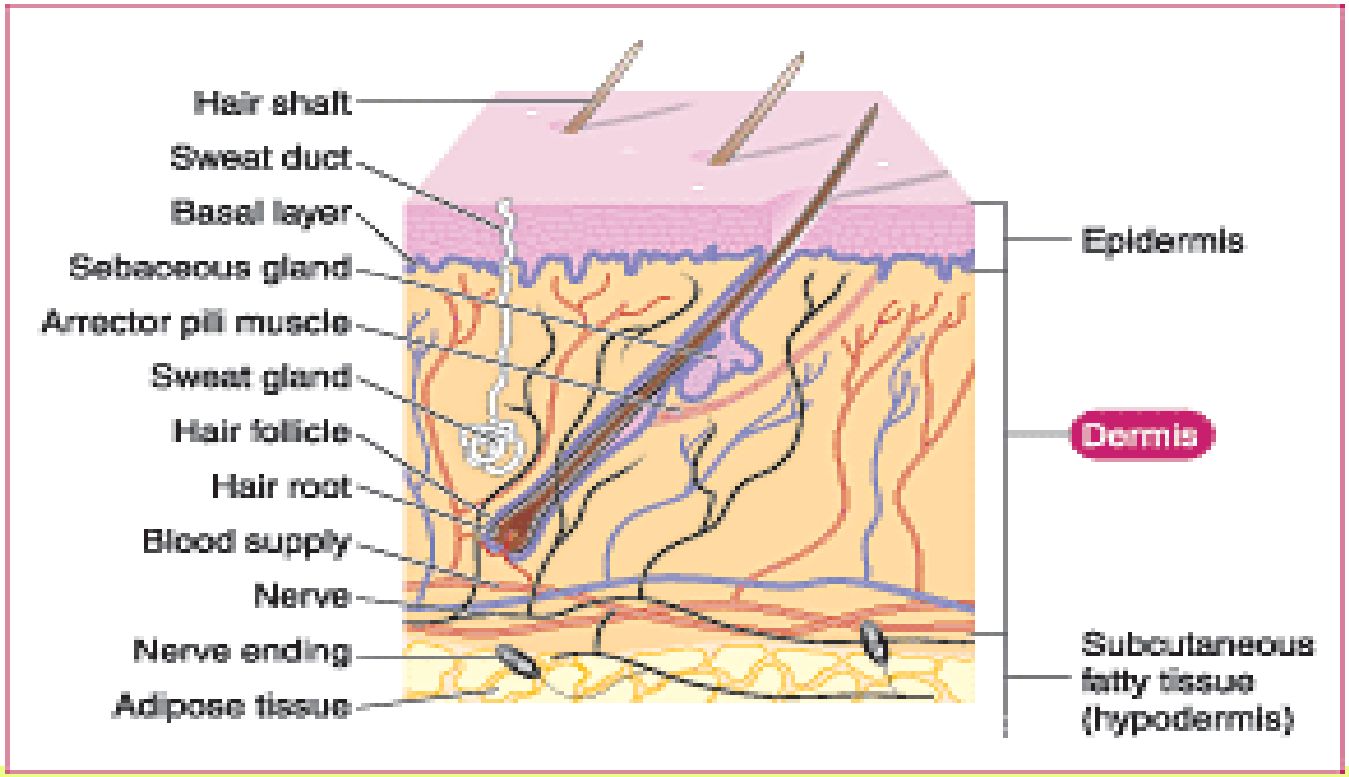
- Blood vessels in the dermis — which is below the basal cell layer — supply nutrients to support this active growth of new skin cells. As the basal cells move upwards and away from their blood supply, their cell content and shape change.

The epidermis



The dermis

- Below the epidermis is the layer called the dermis. The top layer of the dermis — the one directly below the epidermis.



The dermis

1. The dermis contains a variable amount of fat, and also collagen and elastin fibres which provide **strength and flexibility to the skin.**
2. When the skin is exposed to sunlight, **modified cholesterol in the dermis produces vitamin D**, which helps the body to absorb calcium for healthy bones.
3. Blood vessels supply nutrients to the **dividing cells in the basal layer** and **remove any waste products.** They also help maintain body temperature by dilating and carrying more blood when the body needs to lose heat from its surface

The dermis functions

4. Specialised nerves in the dermis(receptors) detect heat, cold, pain, pressure and touch and relay this information to the brain.
5. Hair follicles are embedded in the dermis and occur all over the body, except on the soles, palms and lips.
6. A sebaceous (‘oil’) gland opens into each hair follicle and produces sebum, a lubricant for the hair and skin that helps repel water, damaging chemicals and microorganisms (‘germs’).

The dermis functions

7. Attached to each hair follicle are small erector pili muscle fibres. These muscle fibres contract in cold weather and sometimes in fright .
8. Sweat glands occur on all skin areas — each person has more than 2 million. When the body needs to lose heat these glands produce sweat (a mix of water, salts and some waste material such as urea). Sweat moves to the skin's surface via the sweat duct, and evaporation of this water from the skin has a cooling effect on the body.

Subcutaneous fat (hypodermis)

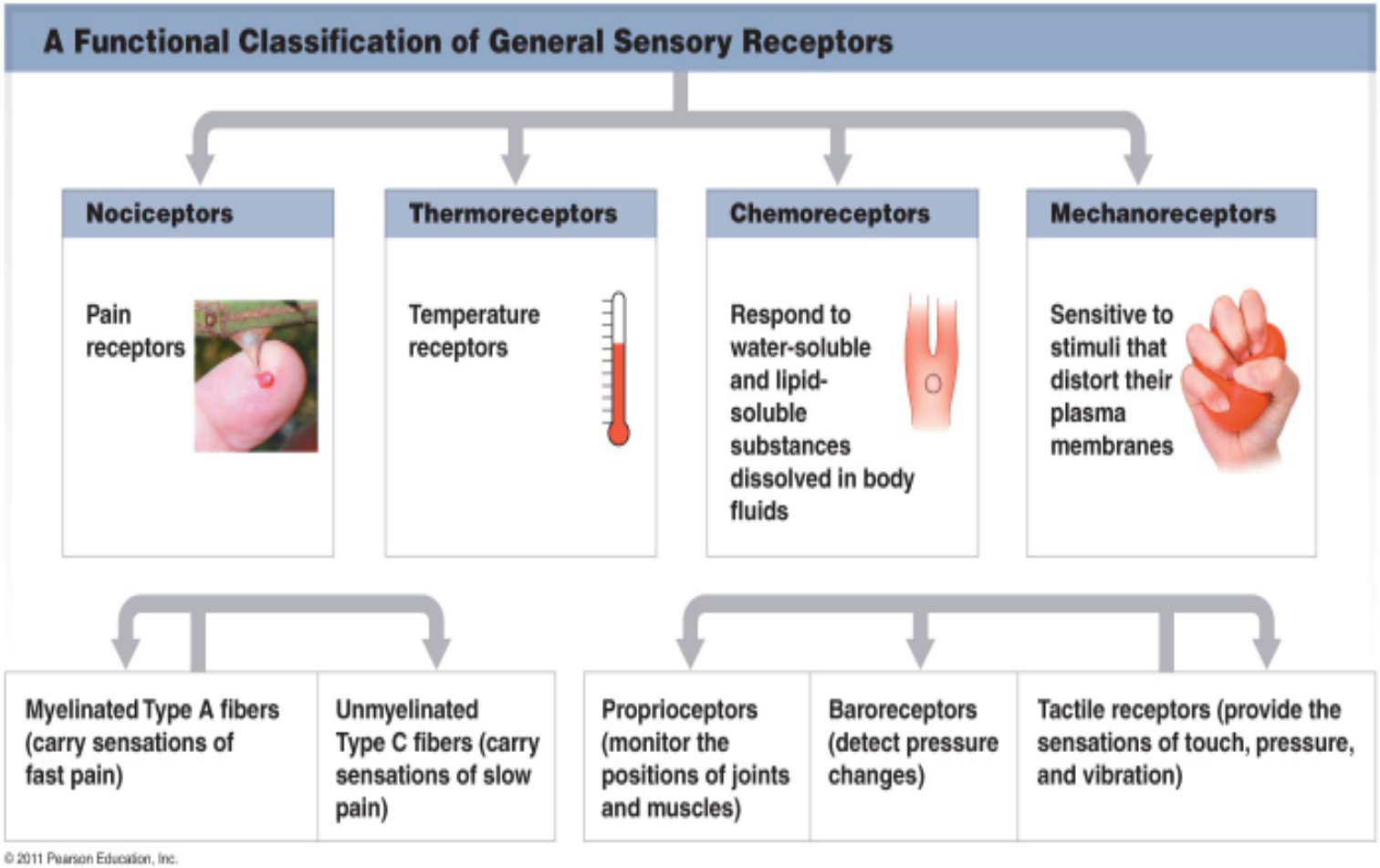
- The innermost layer of the skin is the layer of subcutaneous fat, and its thickness varies in different regions of the body. The fat stored in this layer represents an energy source for the body and helps to insulate the body against changes in the outside temperature.

Receptors

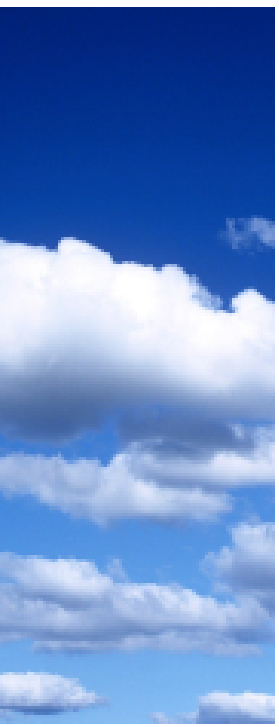
- Receptors are distributed throughout the body as opposed to being concentrated at small, specialized locations
- Responds to many kinds of stimuli

Types of somatic sensation receptors

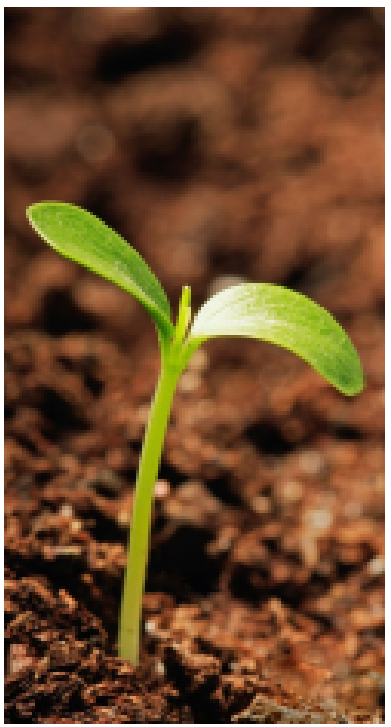
1. Mechanoreceptors--sensitive to physical distortion
2. Nociceptors--respond to damaging stimuli
3. Thermoreceptors--sensitive to changes in temperature
4. Proprioceptors--monitor body position
5. Chemoreceptors--respond to certain chemicals.
6. Photoreceptores: like eye

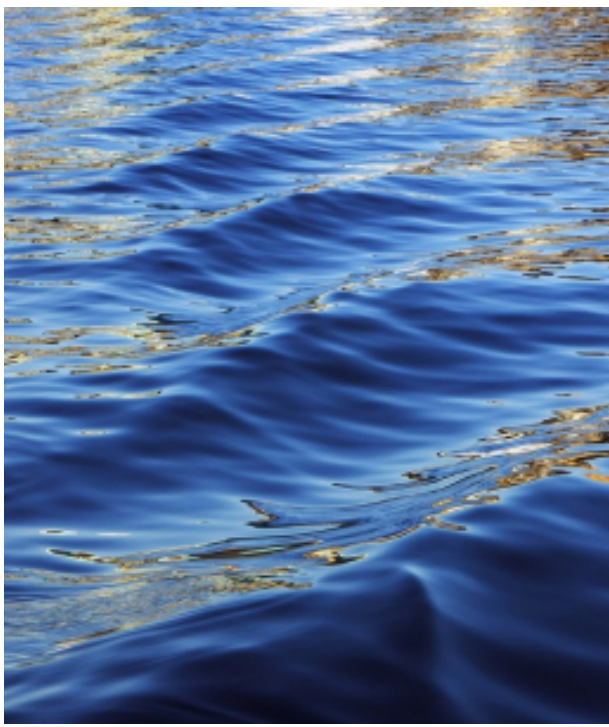


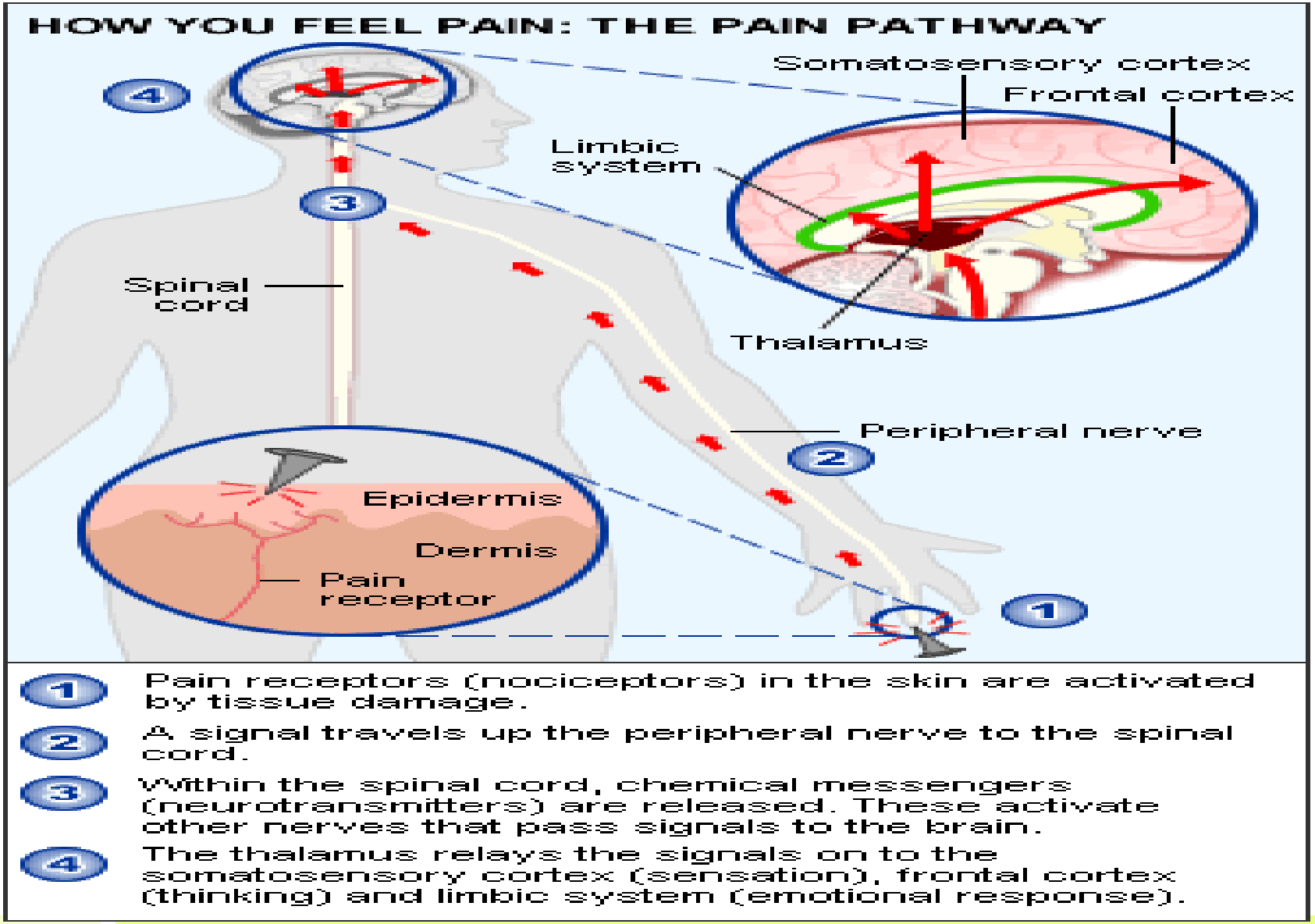
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Now we have to know how receptor signal travel through our body?






HOW YOU FEEL PAIN: THE PAIN PATHWAY
Somatosensory cortex
Frontal cortex
Limbic system
Thalamus
Spinal cord
Peripheral nerve
Epidermis
Dermis
Pain receptor
1
2
3
4
1 Pain receptors (nociceptors) in the skin are activated by tissue damage.
2 A signal travels up the peripheral nerve to the spinal cord.
3 Within the spinal cord, chemical messengers (neurotransmitters) are released. These activate other nerves that pass signals to the brain.
4 The thalamus relays the signals on to the somatosensory cortex (sensation), frontal cortex (thinking) and limbic system (emotional response).

Anatomy and physiology of nerves

- The human body is made up of trillions of cells. Cells of the nervous system, called nerve cells or neurons, are specialized to carry "messages" through an electrochemical process.
- The human brain has approximately 100 billion neurons.
- Neurons come in many different shapes and sizes. Some of the smallest neurons have cell bodies that are only 4 microns wide. Some of the biggest neurons have cell bodies that are 100 microns wide.

Neurons

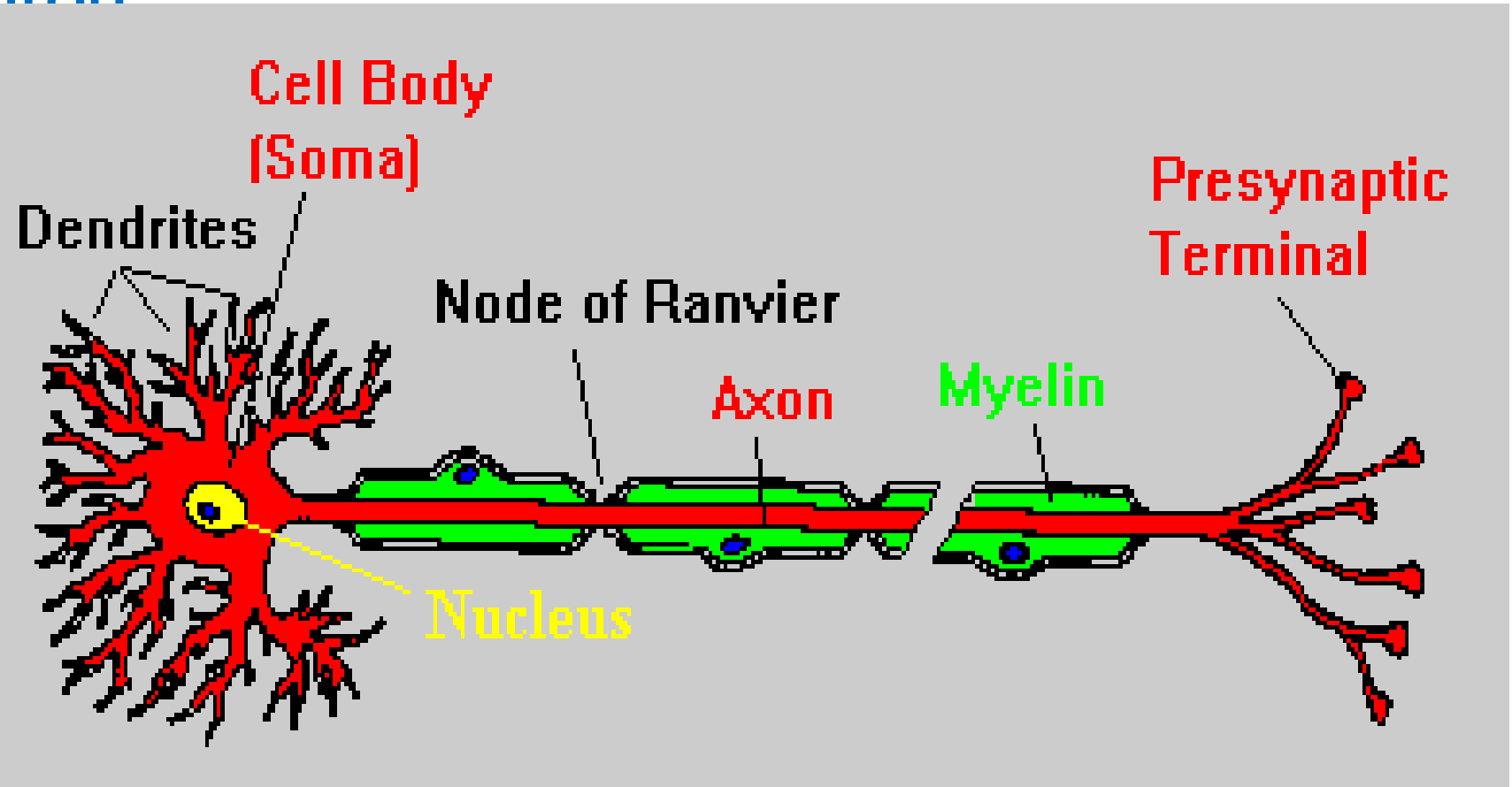
- **Neurons are similar to other cells in the body because:**

1. Neurons are surrounded by a cell membrane.
2. Neurons have a nucleus that contains genes.
3. Neurons contain cytoplasm, mitochondria and other organelles.
4. Neurons carry out basic cellular processes such as protein synthesis and energy production.

Neurons

- **However, neurons differ from other cells in the body because:**
- Neurons have specialize cell parts called **dendrites** and **axons**. Dendrites bring electrical signals to the cell body and axons take information away from the cell body.
- Neurons communicate with each other through an electrochemical process.
- Neurons contain some specialized structures (for example, synapses) and chemicals (for example, neurotransmitters).

neuron

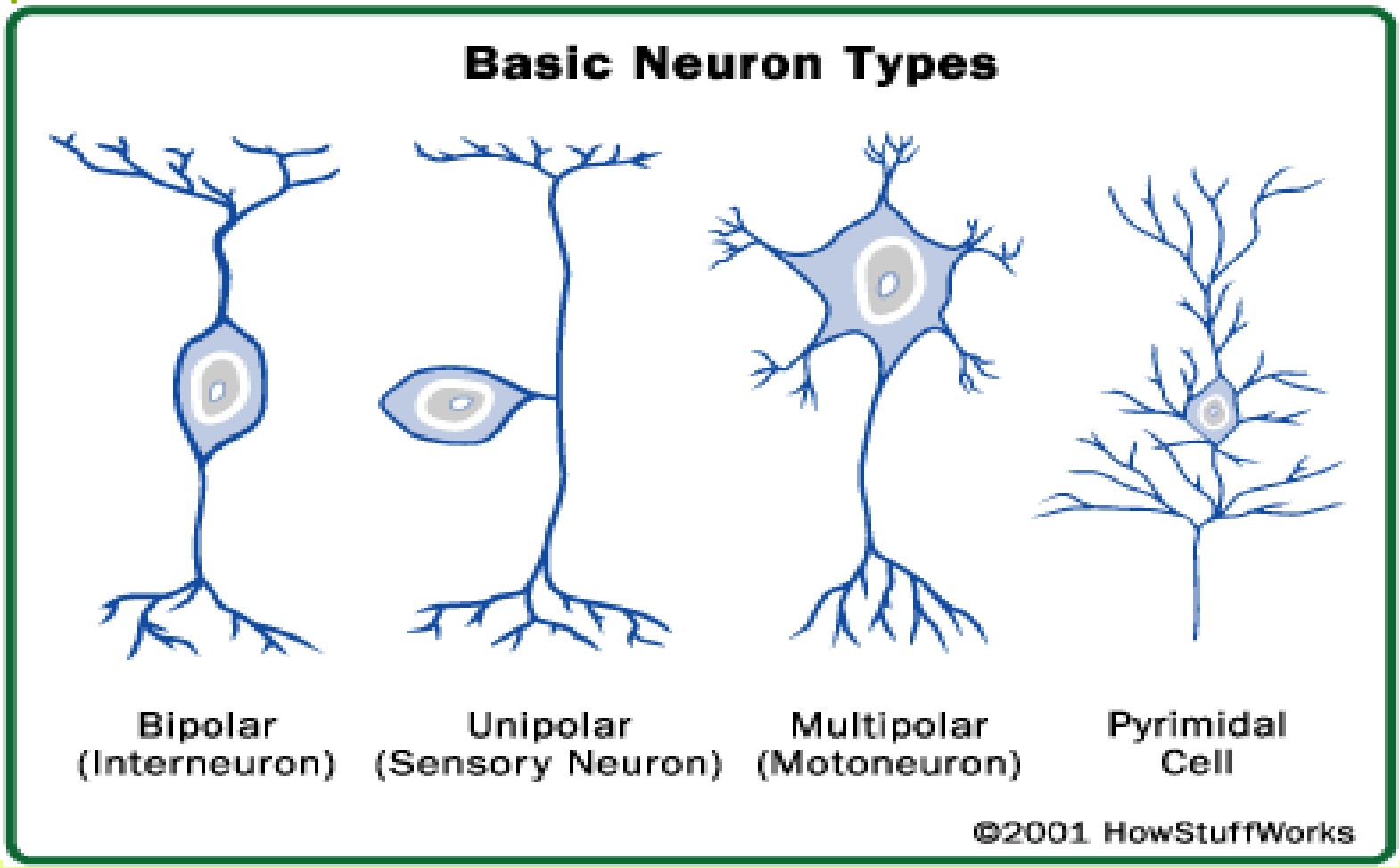

Cell Body
(Soma)
Dendrites
Node of Ranvier
Axon
Myelin
Presynaptic
Terminal
Nucleus


GoAnimate
Go Make Your Own

Types of neuron

- **Bipolar neurons** have two processes extending from the cell body
- **Pseudounipolar cells** (example: dorsal root ganglion cells). Actually, these cells have 2 axons rather than an axon and dendrite. One axon extends centrally toward the spinal cord, the other axon extends toward the skin or muscle.
- **Multipolar neurons** have many processes that extend from the cell body. However, each neuron has only one axon (examples: spinal motor neurons, pyramidal neurons, Purkinje cells).

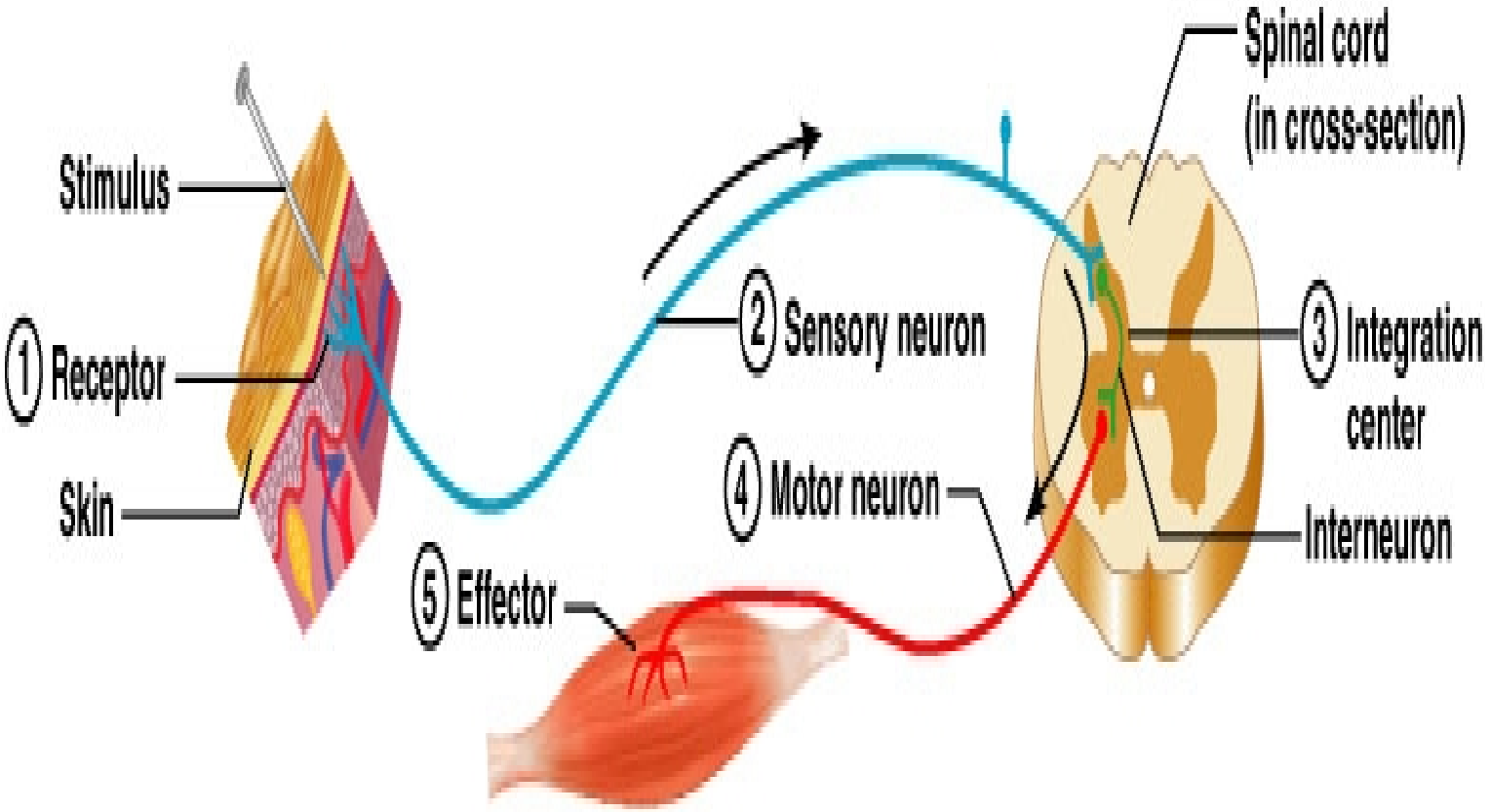
Types of neuron



Classification of neurons

Neurons can also be classified by the direction that they send information.

1. **Sensory (or afferent) neurons:** send information from sensory receptors (e.g., in skin, eyes, nose, tongue, ears) **toword** the central nervous system.
2. **Motor (or efferent) neurons:** send information **away** from the central nervous system to muscles or glands.
3. **Interneurons:** send information between sensory neurons and motor neurons. Most interneurons are located in the central nervous system.



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Nerve fiber

- According to the size and speed of conduction, the nerve fibers are classified into three different categories, namely A, B and C.

Type A fibers

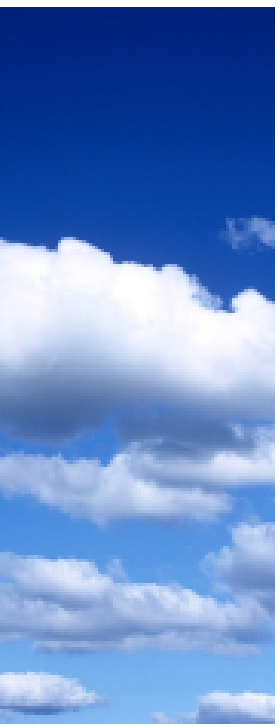
- These fibers are the thickest and fastest conducting.
- They are myelinated.
- They have a diameter of 1.5-20 micron
- Their speed of conduction is 4-120 m/sec, which shows that they have a really fast conduction of impulse.
- Examples of type A fibers are skeletomotor fibers, fusimotor fibers and afferent fibers to skin.

Type B fibers

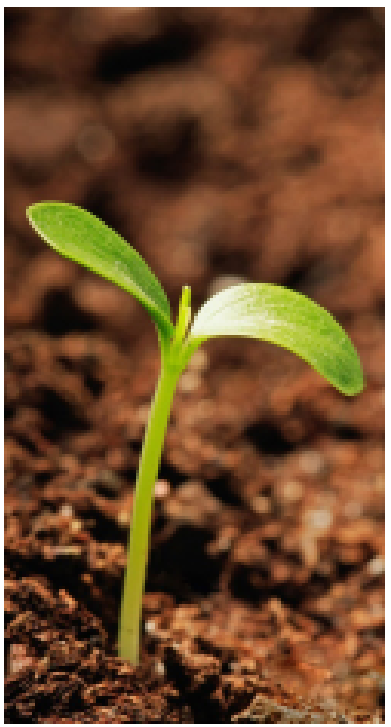
- These fibers are medium in size, i.e. they are smaller than type A fibers but larger than type C.
- They are myelinated.
- They have a diameter of 1.5–3.5 microns.
- Their speed of conduction is 3–15 m/sec, which shows that they are slower than type A fibers.
- Examples of type B fibers are preganglionic autonomic efferents.

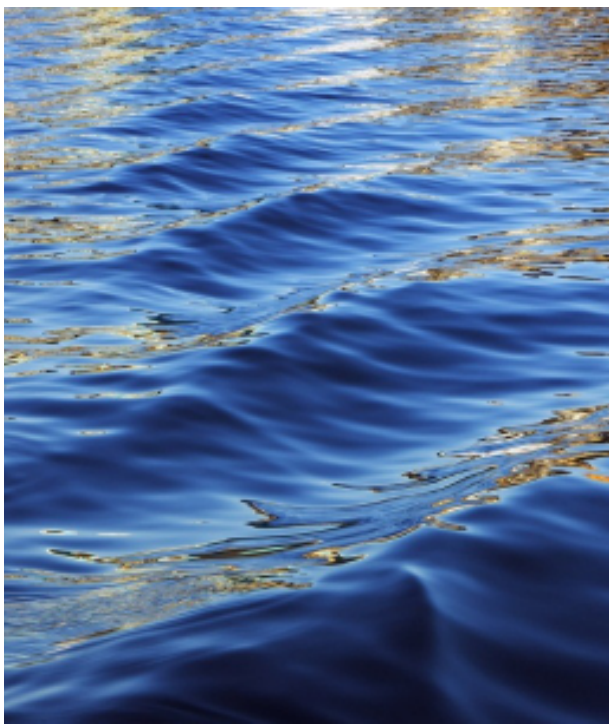
Type C fibers

- These fibers are the smallest and thinnest.
- They are non-myelinated.
- They have a diameter of 0.1-2 microns.
- Their speed of conduction is 0.5-4 m/sec, which shows that they have the slowest conduction.
- Examples of type C fibers are postganglionic autonomic efferents and afferent fibers to skin.



Action potential





Steps of action potential

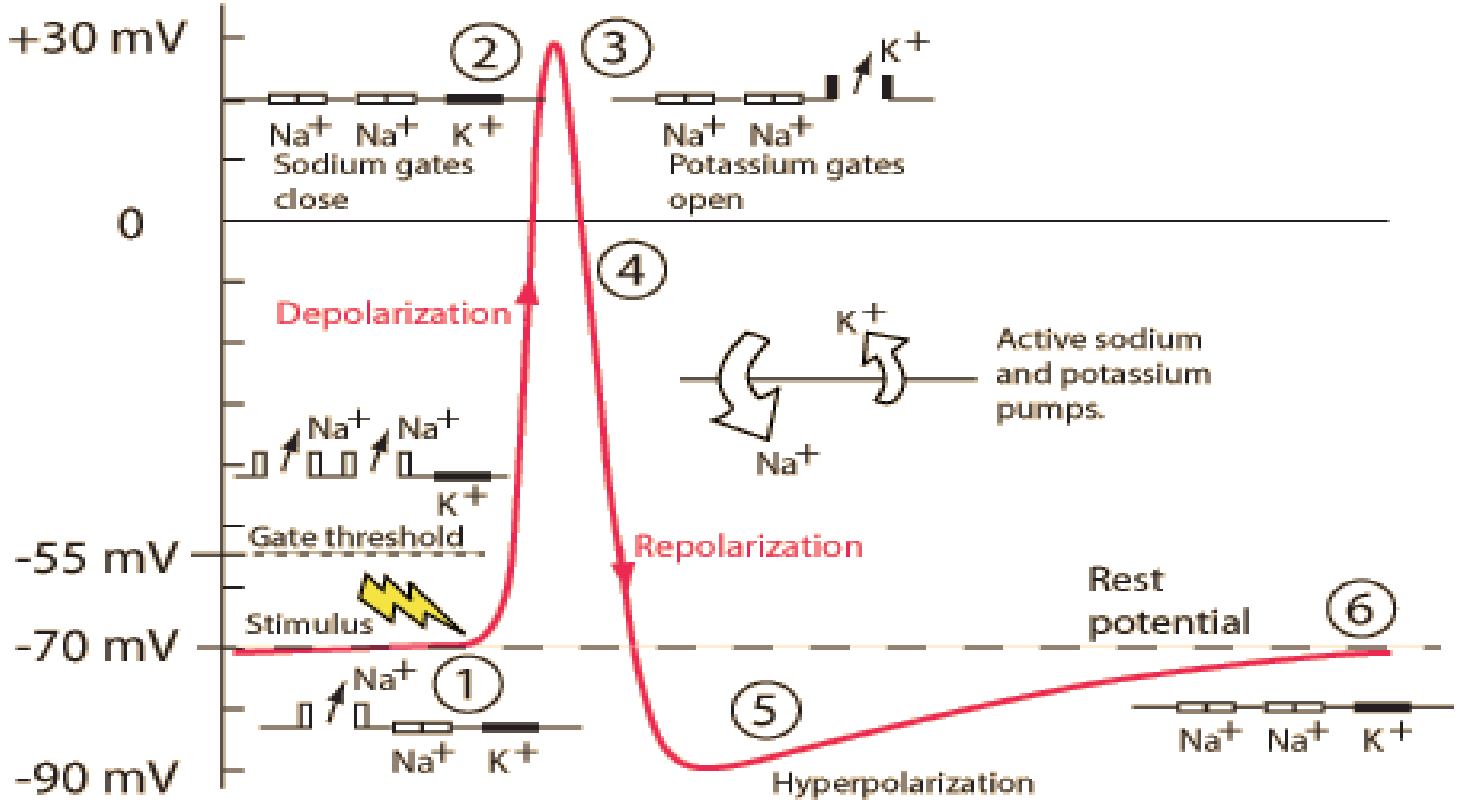
1. A stimulus is received by the dendrites of a nerve cell. This causes the Na^+ channels to open. If the opening is sufficient to drive the interior potential from –70 mV up to –55 mV, the process continues.
2. Having reached the action threshold, more Na^+ channels (sometimes called voltage-gated channels) open. The Na^+ influx drives the interior of the cell membrane up to about +30 mV. The process to this point is called depolarization.

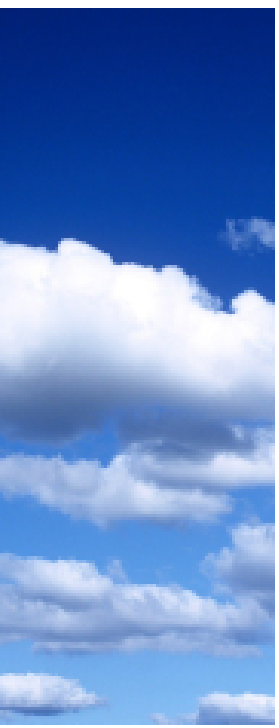
3. The Na^+ channels close and the K^+ channels open. Since the K^+ channels are much slower to open, the depolarization has time to be completed. Having both Na^+ and K^+ channels open at the same time would drive the system toward neutrality and prevent the creation of the action potential.
4. With the K^+ channels open, the membrance begins to repolarize back toward its rest potential.

follow

5. The repolarization typically overshoots the rest potential to about –90 mV. This is called hyperpolarization and would seem to be counterproductive, but it is actually important in the transmission of information. Hyperpolarization prevents the neuron from receiving another stimulus during this time, or at least raises the threshold for any new stimulus. Part of the importance of hyperpolarization is in preventing any stimulus already sent up an axon from triggering another action potential in the opposite direction. In other words, hyperpolarization assures that the signal is proceeding in one direction.
6. After hyperpolarization, the $Na^{\pm}/K^{\pm}$ pump eventually brings the membrane back to its resting state of –70 mV .

Steps of action potential





Thank you

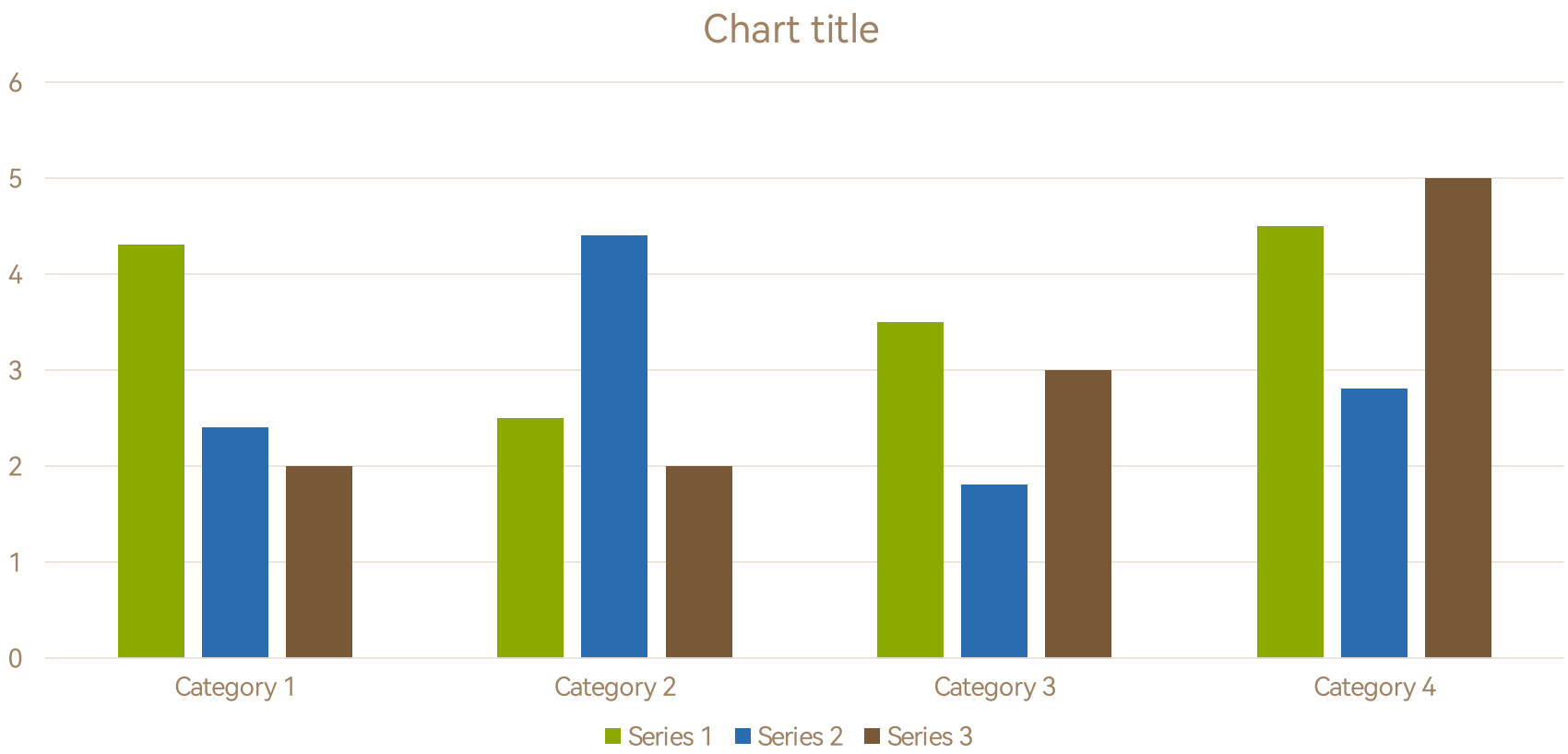




Title and Content Layout with List

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Title and Content Layout with Chart



Two Content Layout with Table

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	Group A	Group B
Class 1	82	95
Class 2	76	88
Class 3	84	90

Two Content Layout with SmartArt

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